

Technology Stack (Architecture & Stack)

Field	Details
Date	02 July 2026
Team ID	Individual Project
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites
Maximum Marks	4 Marks

TECHNOLOGY STACK

Project: Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites
The following technologies are used to develop, deploy and run the application.

1. FRONTEND (User Interface)			
Technology		Purpose	Details
	HTML5	Structure of the web pages	Used to build the structure and layout of the application pages.
	CSS3 / Bootstrap 5	Styling and responsive design	Used for styling, layout, responsive design and UI components.
	JavaScript (ES6)	Client-side interactivity	Used to handle UI interactions, filters, and dynamic content updates.
	Chart.js / Plotly.js	Additional visualizations	Used for interactive charts (if required alongside Tableau).
	Tableau (Embedded)	Interactive dashboards & story	Tableau dashboards and stories are embedded in the Flask application.
2. BACKEND (Server Side)			
Technology		Purpose	Details
	Python 3.x	Backend programming language	Core language used to build the backend logic and application.
	Flask	Web framework	Lightweight Python web framework used to build the web application and REST endpoints.
	Pandas	Data manipulation	Used for data cleaning, transformation and analysis.
	NumPy	Numerical operations	Used for numerical calculations and handling arrays.
	SQLAlchemy	ORM (Object Relational Mapping)	Used for database operations and interaction.
3. DATABASE			
Technology		Purpose	Details
	MySQL	Database management system	Used to store application data such as users (if any), logs, and configuration.
4. DATA SOURCES			
Technology / Source		Purpose	Details
	CSV Files	Raw data source	UNESCO World Heritage Sites dataset (CSV format) used for analysis and visualization.
5. DEPLOYMENT & INFRASTRUCTURE			
Technology		Purpose	Details
	Heroku	Cloud platform	Used to deploy and host the Flask web application.
	Git & GitHub	Version control	Used for source code management and collaboration.
	Nginx	Web server / Reverse proxy	Used to serve the application and handle requests efficiently (via Heroku router or custom setup).
6. TOOLS & UTILITIES			
Technology		Purpose	Details
	VS Code	IDE / Code editor	Used for development and debugging.
	Postman	API testing	Used to test REST APIs (if any).
	Tableau Public	Publishing dashboards	Used to publish dashboards publicly (if required).